Fatima Wasim

Data Scientist · AI Engineer · Analytics & Data Engineering

Sharjah, UAE · UAE Residence Visa, no relocation needed · UAE Driving Licence · Available January 2027
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Portfolio: fatimawasim.github.io

portfolio

PROFILE

Data and AI engineer completing an M.S. in Data Science at the University of Maryland (Dec 2026, GPA 3.88). I have worked across the whole data path: SQL and SSIS pipelines with Power BI reporting at a U.S. commercial bank, an ML model tuned for ~$28M in measured business impact, a cleaning pipeline over 200,000+ customer reviews for a BP-owned travel network, a real-time fraud-detection stream on AWS, and an LLM application on the Claude and Gemini APIs built with explicit limits on what the AI may claim. Raised in Sharjah and back in the UAE on a residence visa. English (fluent), Arabic (basic working).

EDUCATION

M.S. in Data Science, University of Maryland, College Park, USA — Sept 2025 – Dec 2026 (expected)
College of Computer, Mathematical & Natural Sciences · GPA 3.88 / 4.0

B.S. in Computer Science, Youngstown State University, Ohio, USA — Aug 2021 – May 2025
Minor in Mathematics & Data Analytics (Honours) · GPA 3.91 / 4.0 · Summa Cum Laude

High School Diploma (British Curriculum), International School of Creative Science, Sharjah, UAE — June 2021
BEAM High Achiever Award

EXPERIENCE

Data Engineering Intern, First National Bank (FNB), Pennsylvania, USA — May 2024 – Aug 2024
U.S. commercial bank

- Ran the bank's in-house Business Intelligence request platform end to end. I took cross-department requests from scoping through development to production, and delivered Power BI dashboards and SQL/SSMS KPI reports that executive leadership adopted.
- Improved an existing ML classification pipeline through feature engineering and validation on live and held-out data. The pipeline contributed to ~$28M in measured business impact.
- Maintained and optimised SSIS ETL pipelines that move data between internal platforms, keeping the data exchange accurate and secure.
- Took part in Enterprise Data Management Group (EDMG) reviews of data specifications, privacy compliance and schema standardisation.
- Served as Scrum Master for Agile sprints and cleared blockers to keep delivery 100% on time.

Undergraduate Data Science Researcher, The Data Mine (Purdue University) × TravelCenters of America (a BP company), USA — Sept 2024 – May 2025

- Built a Python pipeline that cleaned, validated and restructured 200,000+ raw Yext customer reviews into dashboard-ready data for LLM-based sentiment analysis and prompt-engineering workflows.
- Developed a Kanban-style "Business Action" component (React, TypeScript, MUI) so site managers could track how issues raised in reviews were resolved, including recovery of deleted actions. Also built a Chart.js radar chart that benchmarks each site against its competitors.
- Owned the team's GitHub integration and merge-conflict resolution. Presented the benchmarking findings and recommended actions to executives at company headquarters.

Research Assistant & Student Assistant, Youngstown State University, Ohio, USA — Apr 2022 – May 2025

- Research (Jan 2023 – May 2025): supported graduate research on human behaviour in autonomous-vehicle traffic. I designed heads-up display (HUD) concepts for the study and synthesised literature on brain-signal and eye-movement measures.
- Quality assurance (Apr 2022 – Jan 2024): audited undergraduate and graduate syllabi against institutional compliance criteria, tracked findings in an Excel compliance tracker, and coordinated corrections with faculty and department leadership.
- Co-developed a coding curriculum for K–12 students with Autism Spectrum Disorder with the STEM college and the Computer Science & Information Systems department. Lessons were revised using structured student feedback collected after each session.

SELECTED PROJECTS

EcoJustice MD, Anthropic × UMD Hackathon April 2026

Node.js, Express, Supabase, Claude API, Gemini API, EPA ECHO API

- Built a full-stack civic AI tool that turns a resident's plain-language environmental report into a routed regulatory case. A multi-step LLM workflow classifies the issue, estimates permit likelihood and routes it to the right authority through REST APIs.
- Designed a corroboration layer that merges anonymised witness testimonies into one structured complaint citing the Maryland Environmental Justice Act (2021). Set explicit limits on what the model may assert, and presented the tool to startup CEOs.

Real-Time Credit Card Fraud Detection on AWS, University of Maryland Sept 2026 – Present

EC2, Kinesis Data Streams, Lambda, SageMaker, DynamoDB, SNS, API Gateway

- Built a streaming pipeline: an EC2 transaction generator feeds Kinesis, Lambda calls a SageMaker endpoint serving a fraud classifier I trained, and transactions and flags are stored in DynamoDB.
- Added SNS alerts for flagged transactions and an API Gateway-backed live dashboard. Now measuring accuracy, latency and cost.

Time-Aware Next-Song Prediction with a Transformer (HSTU), University of Maryland Sept 2026 – Present

PyTorch, RecTools

- Implementing a time-aware generative recommender on 19M Last.fm listening events, benchmarked against SASRec, BERT4Rec and baselines on HR@10 and NDCG@10. Verified the implementation first by reproducing published HSTU results on MovieLens-1M.

Chronic Absenteeism Forecasting in U.S. Public Schools, Senior Capstone, YSU Jan 2025 – May 2025

Python, Prophet, ARIMA, Random Forest, regression

- Combined millions of student attendance records across all 50 states (2017–2023) from federal and state sources, reconciling inconsistent metrics and missing data through cross-source validation. Forecast recovery trends and presented the findings to department heads.

TECHNICAL SKILLS

Programming & Data: Python (pandas, NumPy, scikit-learn), SQL (SSMS, relational databases), R, Stata, JavaScript/TypeScript

Machine Learning & AI: PyTorch, TensorFlow, Hugging Face, transformer models, recommender systems, time-series forecasting (Prophet, ARIMA), LLM integration (Claude API, Gemini API), prompt engineering, multi-step LLM workflows, RAG & embeddings (coursework), responsible AI

Data Engineering & Cloud: AWS (EC2, Kinesis, Lambda, SageMaker, DynamoDB, SNS, API Gateway), SSIS & ETL design, data cleaning & validation, REST APIs, Supabase, Linux, Microsoft Azure (coursework), Azure DevOps

BI & Analytics: Power BI (dashboards, KPI design), Power Query, Tableau, Excel (PivotTables), SharePoint, competitive benchmarking

Delivery & Governance: Data governance, privacy compliance reviews, Agile/Scrum (Scrum Master), Jira, Confluence, Git/GitHub, React, Node.js/Express

HONOURS, LEADERSHIP & LANGUAGES

Honours: Clarence P. Gould Award (top 5%) · Honors Scholar with Distinction (Medallion & Pin) · International Scholar Award · Mathematics Department Award · President's List ×6

Leadership: Vice President, Data Science Club (YSU) · Association for Women in Mathematics · STEM Leadership Society · Data Science Salon, Oracle HQ, Austin (2024)

Languages: English (fluent) · Arabic (basic working knowledge)